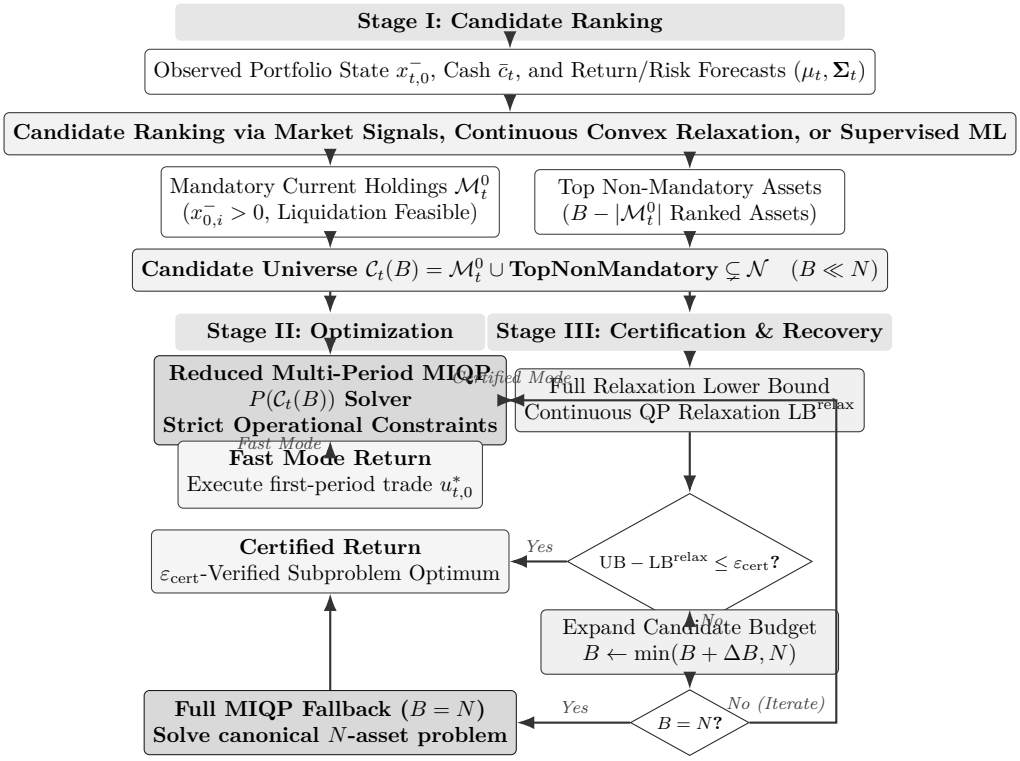

Figure 1: Architecture of the candidate-reduced repeated MIQP framework. Stage I: Statistical ranking or continuous convex relaxations propose candidate sets while mandatory inclusion preserves state continuity and liquidation feasibility. Stage II: A fully constrained mathematical MIQP solver enforces all cardinality limits, transaction costs, and minimum charged trade floors over the reduced subproblem. Stage III: An a posteriori continuous relaxation lower bound certifies subproblem optimality or triggers systematic budget expansion up to full-universe fallback.